06 The Migration Room

Gene flow against drift

CORE PATH · BSC	EXTENSION · MSC	BEFORE THIS ROOM
about 35 min	**+ 30 min**	The Drift Room (04), and the Effective Size Room (05) for what *N* means here. Step 6 needs the Selection Room (07): come back for it.
steps 1–4	steps 5–7	

OPEN THE ROOM

`https://evolinus.github.io/zooe/teaching/evolutionary-laboratories/#migration`

Two populations, A and B, each of *N*, drift on their own each generation and then exchange *M* individuals in each direction. This guide asks how much exchange it takes to stop two populations becoming different, and what *different* should mean.

LEARNING OBJECTIVES

After the core path you should be able to

- Explain why two isolated populations drift apart and often fix different alleles.
- Show that a small, steady exchange of migrants holds two populations together.
- Use F_{ST} as a measure of differentiation, and read it from the room.
- Explain why the number of migrants, Nm, matters more than the migration rate m.

The extension adds

- Interpret the sweep: the mean F_{ST} against the share of runs that ever become strongly differentiated.
- Compare the room's results with $F_{ST} \approx 1/(1 + 4Nm)$ and with the version for two haploid populations, $1/(1 + 8Nm)$, and say why the first does not fit and where the second stops fitting.
- Explain migration–selection balance: why local adaptation survives when $m < s$ and is swamped when $m > s$.

SUGGESTED ROUTES

- **40-minute practical (BSc):** steps 1–4 and questions 1–3.
- **Conservation-themed session:** steps 2–4 with question 3, as an argument about corridors between reserves.
- **MSc:** steps 5–7 and questions 4–6 (about 35 minutes; the sweep takes a little while to run). Step 6 switches selection on, so leave it until after the Selection Room (07).

How to work through it. Every step asks you to *predict* before you press anything, and to *explain* afterwards. Write the prediction down and commit to it: a wrong prediction you can account for teaches more than a right one you guessed. Runs are random, so where a step says so, run it more than once before you decide what happened.

1 One migrant a generation

8 min

SET UP

The defaults: $N = 50$, A starts at $p_A = 0.90$, B at $p_B = 0.10$, 1 migrant each way, $G = 200$.

PREDICT FIRST

Population B starts with A1 rare. With one migrant crossing each way per generation, what will happen to the two frequencies?

RUN

Press **Run Simulation**. Watch the chart, and the *Crossed this gen.* readout.

LOOK AT

the two lines on *Frequency of A1 in each population*, and how the run ends.

EXPLAIN

Why do the two populations come together rather than drifting apart? How do they end up?

2 No migrants

8 min

SET UP

Migrants each way: **0**. Everything else as in step 1.

PREDICT FIRST

What will the two lines do now? In how many of three runs will the two populations end up fixed for different alleles?

RUN

Run three times.

EXPLAIN

With no gene flow, each population is a separate Drift Room. Why does that so often end in A and B being completely different?

3 Twenty replicates

12 min

SET UP

Both starting frequencies p = **0.50**. Migrants 0, then 1.

PREDICT FIRST

Out of twenty runs, how many will at some point become strongly differentiated ($F_{ST} \geq 0.5$) with no migrants? With one migrant?

RUN

Press **Run 20 Replicates** at 0 migrants, then at 1.

LOOK AT

the *Twenty replicates* panel and its count of differentiated runs.

EXPLAIN

How big a difference did one migrant a generation make? Why does the room insist you look at many runs and not one?

4 Count, not fraction

7 min

SET UP

Migrants: 1. N = **20**, then N = **200** (both starting at p = 0.5).

PREDICT FIRST

At N = 20, one migrant is 5% of the population; at N = 200 it is 0.5%. Will the larger populations, with ten times less migration as a *rate*, become more differentiated?

RUN

Run 20 Replicates at each N and note the count of differentiated runs, and the readouts of m and Nm in the deck.

EXPLAIN

What stayed the same between the two settings? Why does a smaller rate in a larger population hold it together just as well?

5 The sweep

12 min

SET UP

$N = 50$, $G = 200$. Nothing else matters: the sweep starts every run at $p = 0.5$.

PREDICT FIRST

Sketch two curves against Nm from 0 to 10: the mean F_{ST}, and the share of runs that ever reach $F_{ST} = 0.5$. Where, if anywhere, do you expect a sharp drop?

RUN

Press **Run the sweep** and wait for the panel to fill.

LOOK AT

the teal mean and the gold share.

EXPLAIN

Which curve has the cliff, and where? Calculate $1/(1 + 4Nm)$ and $1/(1 + 8Nm)$ at $Nm = 1$ and compare with the mean you measured. Why does a model of two populations bind them more tightly than the many-island model behind the textbook formula, and what changes because the individuals here are haploid?

6 Local adaptation against gene flow

10 min

SET UP

Selection: **Opposite in A and B**, $s = 0.10$, $N = 50$, both starting at $p = 0.5$. Migrants 1 ($m = 0.02$), then 10 ($m = 0.2$).

PREDICT FIRST

In which setting will each population keep its own favoured allele? Use the comparison of m with s.

RUN

Run each setting twice.

EXPLAIN

Explain *gene swamping*. Why can immigrants from elsewhere prevent a population from adapting to where it lives?

7 Mutation between two alleles

8 min

SET UP

Selection: None, p = 0.5 in both, 0 migrants, Mutation rate μ = **0.005**, G = 1000.

PREDICT FIRST

Will mutation keep the populations differentiated for longer, or make them more alike?

RUN

Run 20 Replicates with μ = 0 and then with μ = 0.005, and compare the mean F_{ST}.

EXPLAIN

With only two alleles, why does mutation work *against* differentiation? Which assumption of the textbook formula does that break?

DISCUSSION AND WORKSHEET QUESTIONS

1. BSc Two nature reserves hold the same butterfly. A corridor would let about two butterflies a generation move between them. Is that worth building, if the populations are 50 each? If they are 5,000 each?
2. BSc "These populations exchange less than 1% of their individuals, so they must be diverging." Use the room to criticise this argument.
3. BSc Why does the room report F_{ST} rather than just the difference in p between the populations?
4. MSc The rule "one migrant per generation" comes from a model with assumptions. List three of them, and say in which direction breaking each would move the threshold.
5. MSc Migrants in nature are rarely a random sample of their population. Give an example and say how it would change step 6.
6. MSc Why does the time to reach equilibrium grow with N, even though the equilibrium itself depends only on Nm?

CHECKING YOUR REASONING

Read this after you have written your own answers. These are what the model should show on most runs — not the only acceptable explanations.

- **Step 1.** B climbs towards A, the two lines stay together, and the pair then loses its variation *as a pair*, usually within the first couple of hundred generations.
- **Step 2.** Without migrants each population drifts on its own; about half the runs end with A and B fixed for different alleles.
- **Step 3.** Expect around thirteen of twenty strongly differentiated with no migrants, and around five with one.
- **Step 4.** The counts are similar: $Nm = 1$ in both, even though m differs tenfold. Larger populations drift more slowly, so they need fewer migrants as a fraction.
- **Step 5.** The mean F_{ST} slides smoothly; the share reaching 0.5 falls off a cliff near $Nm = 1$. At $Nm = 1$ the room measures a mean near 0.09, against 0.2 for the island formula and 0.11 for $1/(1 + 8Nm)$: two populations bind four times more tightly than many islands, and a haploid migrant carries one gene copy where a diploid one carries two. Below one migrant the room falls short of $1/(1 + 8Nm)$, because 200 generations from $p = 0.5$ is not long enough to reach the equilibrium the formula describes.
- **Step 6.** At $m = 0.02 < s$ each population keeps its own allele; at $m = 0.2 > s$ gene flow swamps local adaptation and both end up with the same allele.
- **Step 7.** Mutation lowers the mean F_{ST}: symmetric mutation between two alleles pulls both populations towards 0.5. The textbook formula assumes infinitely many alleles, where mutation never makes two populations alike.